LLM Watermarking

Raj Nadkarni, Sebastian Woo, Leo Li, Ashton Liu

What's Our Project About?

- Detecting whether or not outputs come from a particular LLM can be useful
  - Determining if something was written by a LLM vs actual person
  - Detecting cheating in academics
  - Upholding Copyright and Compliance laws
  - Prevent retraining on generated data (leads to model collapse)
- We can achieve this with LLM watermarking

What Makes A Good Watermark?

Kuditipudi et al. (2023) states that the ideal watermark should satisfy at least the following:

1. Preserve the quality of original text as much as possible
2. Detectable without the language model and/or prompt
3. Withstands perturbations of the watermarked text

https://arxiv.org/pdf/2307.15593

How Our “Green List” Watermark Works

- During inference, we use a keyed PRP to determine a “green list” of tokens
  - LLM samples “green” tokens with higher probability
- For detection, use the key to see if “green” tokens were preferred with statistical significance to determine LLM usage within text
- Let n = number of scored tokens (seqlen - 1), p = the green list proportion aka gamma, X = actual number of green tokens
- X ~ Binomial(n, p)
  - Mean = E[X] = np
  - Var[X] = np(1 - p)
  - Z = (X - np) / sqrt(np(1-p))
- Select some Z-threshold to determine whether or not text is watermarked

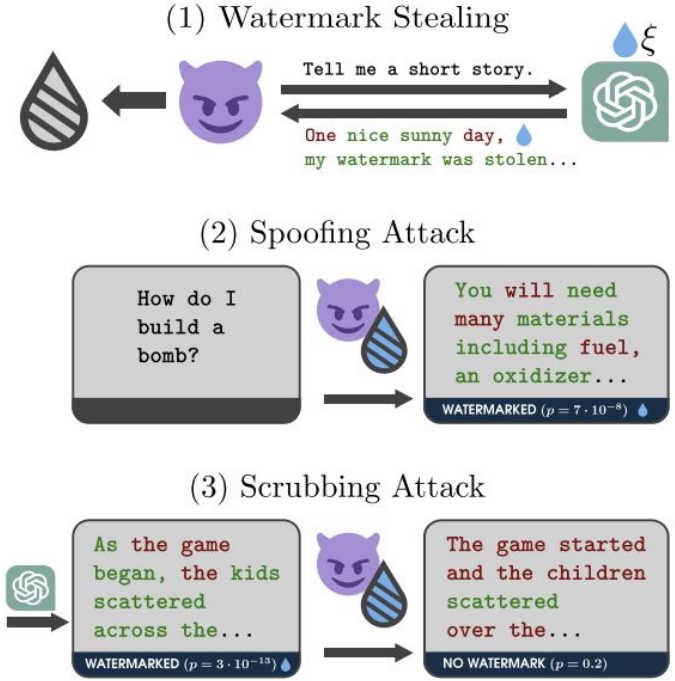
Demo

Considerations

- Aggressive watermarking can hurt the quality of model outputs
  - Certain token sequences that should appear might have a low probability of being generated
  - E.g. “Sherlock Holmes” is a common phrase, but if “Holmes” ends up on the “red list” for “Sherlock”, a high enough gamma will significantly impact generation
- Attacks on watermarks (Diaa et. al)

Attack Name	Description
DIPPER [Krishna et al., 2023]	Train an 11b Sequence-to-Sequence model for paraphrasing.
Translate [Piet et al., 2023]	Translate to another language and back (e.g., French, Russian).
Swap [Piet et al., 2023]	Randomly remove, add or swap words.
Synonym [Piet et al., 2023]	Replace words with a synonym using WordNet [Miller, 1995].
HELM [Bommasani et al., 2023]	Randomly add typos, lowercase or contractions
Llama, Qwen, GPT3.5	Paraphrase text using a publicly accessible LLM.

Advanced Attacks



Jovanovic et al. (2024)

- Demonstrated that many SOTA watermark schemes can be stolen for under $50 in OpenAI API costs
- These allow for other spoofing or scrubbing attacks

References

- https://arxiv.org/pdf/2301.10226
- https://arxiv.org/pdf/2307.15593
- https://arxiv.org/pdf/2402.19361
- https://youtu.be/55LXhbPjyeM